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Research paper

Further results on fixed-time synchronization of the memristor neural networks with impulsive effects

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A B S T R A C T

The article is mainly concerned with the fixed-time synchronization (FXTS) of memristor neural networks (MNNs) with time-varying delay and impulsive effects. Using the concept of the average impulsive interval and the comparison principle, some sufficient conditions have been derived to achieve the FXTS of MNNs under the effects of synchronizing impulses and desynchronizing impulses. For the case of desynchronizing impulses, the FXTS of MNNs have not been investigated in literature. Moreover, it has been shown through numerical simulation and theoretical derivation that the estimated settling time is less conservative than that of previous results for MNNs under the effects of synchronizing impulses. Finally, a numerical example is provided to illustrate the effectiveness and superiority of the improved synchronization methodology.

1. Introduction

The term and concept of memristor, which stands for memory and resistor, was coined by Leon Chua in 1971. It is well known that none of the three fundamental circuit elements viz. resistor, capacitor and inductor describe the relationship between charge and flux. This missing relation between charge and flux is covered by the fourth fundamental circuit element memristor. After the first physical demonstration of memristor at HP Lab in 2008, several models like the Strukov model and Simmon tunnel barrier model have been introduced for different applications [1,2]. Having the property of non-volatility and being a nanoscale element, the memristor is the best electronic component to play a role as synapses to imitate neural networks better. The memristor can be used to replicate the synapsis in the human brain since it has the property of memorizing the direction of the past charge flow. Recently, the memristor has gained more interest because of its essential role in the next generation of computers and powerful human brain-like computers. Thus, the introduction of the memristor in place of the resistor in neural networks opens a new door in the study of the neural networks, known as memristor neural network (MNN), which has a vast field of applications such as in image processing, secure communication, fault diagnosis and pattern recognition. In the past two decades, researchers have made significant progress in their understanding of the dynamical behavior of MNNs, and a wide range of interesting results have been published in the Refs. [3–7].

The term synchronization generally means that as time evolves, the dynamical states of coupled systems tend to coincide. Synchronization is gaining a lot of importance in recent years due to its various applications in engineering fields such as chemical reaction, information processing, secure communication, and biological system [8–10]. There have been many synchronization schemes used theoretically and practically, for instance, exponential synchronization [11],

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lag synchronization [12], complete synchronization [13], projective synchronization [14], phase synchronization [15] and adaptive synchronization [16]. To begin with, synchronization in infinite-time was studied in terms of asymptotic synchronization and exponential synchronization. However, the finite life spans of appliances and biological systems make infinite-time synchronizations impractical in real life. To counter the beforementioned drawbacks, a kind of synchronization known as finite-time synchronization (FTS) has come into practice and is being widely studied in the recent past [12,17–22]. It is worth to be noted that the settling-time in FTS depends heavily on the initial states of the system, which makes the settling-time vary for different initial conditions. Further, not all the initial values of a system in practice are available to be measured. Hence, in order to deal with these kinds of drawbacks, a new type of synchronization concept namely, fixed-time synchronization (FXTS) was proposed in [23]. The upper hand of FXTS over FTS is that its settling time does not depends on the initial conditions and thus has a wide range of applications than that of FTS. Several significant promising studies on FXTS of MNNs have been derived by many researchers [24–30]. For example, in [24] FXTS of a class of delayed MNNs based on Lyapunov functional, analytical techniques and using discontinuous controller have been given. Using Filippov discontinuity theory and Lyapunov stability, the article [25] reported both FTS and FXTS problems of inertial MNNS with time-varying delay and in Ref. [26], the authors discussed FXTS of inertial MNNs with discrete delay. In [27], the FXTS problems for stochastic MNNs were proposed using differential inclusions theory and set-valued maps. The fixed-time stability theorem developed in [23], on which the Lyapunov condition should satisfy the inequality $\dot{W}(t) \leq -aW^{\alpha}(t) - bW^{\beta}(t)$, where $a, b > 0$, $\alpha > 1$ and $0 < \beta < 1$, was the main theory from which all of the above FXTS results and settling-time estimations were obtained. Recently Chen et al. [31] derived a new fixed time stability theorem, in which the Lyapunov condition satisfies the inequality $\dot{W}(t) \leq -aW^{\alpha}(t) - bW^{\beta}(t) - cW(t)$, where $a, b, c > 0$, $\alpha > 1$ and $0 < \beta < 1$. Due to one additional term $cW(t)$, the settling-time estimation in [31] is significantly lesser when one compares it with the previous results in [23].

The state of the neural networks changes abruptly at certain instants of time due to some external disturbances or other sudden noise, which exhibit impulsive effects. In general, there are three categories for impulse strength (η_l) as follows: synchronizing impulses ($|\eta_l| < 1$), desynchronizing impulses ($|\eta_l| > 1$) and inactive impulses ($|\eta_l| = 1$). Therefore, it is very important to study the dynamical behavior of neural networks with impulsive effects. Up to now, a significant number of papers on impulsive effects have been reported in [18,32–35]. However, there are a few articles in which FXTS of neural networks with impulsive effects have been investigated [36–41]. Furthermore, authors have only studied the synchronizing impulsive effects. The desynchronizing impulsive effects have rarely been considered in literature. To the best of our knowledge, there are no relevant results regarding FXTS of MNNs for desynchronizing impulsive effects available in literature. Thus, it is an important and inspiring problem to explore the newly developed fixed-time synchronization theories with lesser estimated settling time under the effects of synchronizing impulses and desynchronizing impulses for MNNs. Hence, we attempt here to investigate the FXTS of MNNs with impulsive effects.

Motivated by the above discussions, in this article, the FXTS of MNNs with impulsive effects and time delays have been investigated. Using the concept of comparison principle and average impulsive interval, we have derived sufficient conditions so that the concerned MNNs will achieve FXTS under the influence of synchronizing impulses and desynchronizing impulses. The main contribution of this article can be described as follows:

1. The comparison principle and average impulsive interval are used to derive sufficient criteria for FXTS of MNNs under the influences of synchronizing impulses and desynchronizing impulses.
2. Both, theoretically and numerically it has been shown that the estimated settling-time is less conservative and of high precision as compared to those results found in the existing FXTS reported in [37–39,42].
3. The FXTS of MNNs for desynchronizing impulses have not been discussed in prior results reported in [37]. In this article, we make attempt to find the condition under which MNNs with desynchronizing impulses can achieve FXTS.

The remaining part of the paper is assembled as follows. Section 2 consists of a detailed description of the model and some important preliminaries, such as definitions, assumptions and lemmas, which are used throughout the paper. In Section 3, the FXTS criteria of MNNs are derived for two classes of impulses (synchronizing and desynchronizing impulses). The efficiency of the theoretical results of this article is verified in Section 4 with the help of numerical examples of MNNs. The overall conclusion of the paper is presented in Section 5.

Notations. Throughout the paper, $\mathbb{R}$, $\mathbb{R}_+$ and $\mathbb{R}^n$ denote the set of the real number, the set of a non-negative real numbers and the n-dimensional Euclidean space, respectively. $\mathbb{N}$ is the set of natural numbers.

2. Model description and preliminaries

Let us consider the state equation of MNNs with impulsive effects as

$$\begin{cases} \dot{z}_p(t) = -d_p z_p(t) + \sum_{q=1}^{n} h_{pq}(z_p(t))f_q(z_q(t)) + \sum_{q=1}^{n} w_{pq}(z_p(t))g_q(z_q(t-\tau_q(t))) + I_p, t \geq 0, \quad t \neq t_l, \\ \Delta z_p(t_l) = \eta_l z_p(t_l^-), t = t_l, l \in \mathbb{N}, \end{cases} \tag{1}$$

where $p = 1, 2, \ldots, n$, $z_p(t) \in \mathbb{R}$ represent the voltages of capacitors; $d_p > 0$ are the self-inhibition; $f_q(.)$ and $g_q(.)$ denote the activation functions of the qth neuron; $\tau_q(t)$ corresponds to the time-varying delay; $h_{pq}(.)$ and $w_{pq}(.)$ represent memristor-based connection weights; I_p denotes the external input or bias. The sequence $\{t_1, t_2, t_3 \ldots\}$ is strictly increasing impulsive instants, η_l denotes the impulsive strength, $\Delta z_p(t_l) = z_p(t_l^+) - z_p(t_l^-)$, where $z_p(t_l^+) = \lim_{t\to t_l+0} z_p(t)$ and $z_p(t_l^-) = \lim_{t\to t_l-0} z_p(t)$. Suppose $z_p(t)$ is right continuous at $t = t_l$, i.e., $z_p(t_l^+) = z_p(t_l)$. The initial condition of (1) is $z_p(s) = \phi_p(s)$, $s \in [-\tau, 0]$. The memristor-based connection weights in (1) satisfy

$$h_{pq}(z_p(t)) = \begin{cases} \hat{h}_{pq}, & |z_p(t)| \le T_p, \\ \check{h}_{pq}, & |z_p(t)| > T_p, \end{cases} \qquad w_{pq}(z_p(t)) = \begin{cases} \hat{w}_{pq}, & |z_p(t)| \le T_p, \\ \check{w}_{pq}, & |z_p(t)| > T_p, \end{cases}$$

for $p, q = 1, 2, \dots, n$, where $\hat{h}_{pq}$, $\check{h}_{pq}$, $\hat{w}_{pq}$, $\check{w}_{pq}$ are known constants with respect to memristances, and T_p are non-negative threshold level. According to the theory of differential inclusion and set valued map [43,44], it can be obtained that the system (1) is equivalent to the following differential inclusion

$$\dot{z}_p(t) \in -d_p z_p(t) + \sum_{q=1}^{n} K(h_{pq}(z_p(t))) f_q(z_q(t)) + \sum_{q=1}^{n} K(w_{pq}(z_p(t))) g_q(z_q(t-\tau_q(t))) + I_p, t \neq t_l, \tag{2}$$

where

$$K(h_{pq}(z(t))) = \begin{cases} \hat{h}_{pq}, & |z(t)| < T_p, \\ \mathrm{co}\{\hat{h}_{pq}, \check{h}_{pq}\}, & |z(t)| = T_p, \\ \check{h}_{pq}, & |z(t)| > T_p, \end{cases} \quad K(w_{pq}(z(t))) = \begin{cases} \hat{w}_{pq}, & |z(t)| < T_p, \\ \mathrm{co}\{\hat{w}_{pq}, \check{w}_{pq}\}, & |z(t)| = T_p, \\ \check{w}_{pq}, & |z(t)| > T_p, \end{cases}$$

where $\mathrm{co}\{a, b\}$ denotes the convex closure of a set. According to the measurable selection theorem, there exists measurable function $h^*_{pq}(z_p(t)) \in K(h_{pq}(z_p(t)))$ and $w^*_{pq}(z_p(t)) \in K(w_{pq}(z_p(t)))$ such that

$$\begin{cases} \dot{z}_p(t) = -d_p z_p(t) + \sum_{q=1}^{n} h^*_{pq}(z_p(t)) f_q(z_q(t)) + \sum_{q=1}^{n} w^*_{pq}(z_p(t)) g_q(z_q(t-\tau_q(t))) + I_p, \quad t \neq t_l, \\ z_p(t_l) = (1+\eta_l) z_p(t_l^-), t = t_l, l \in \mathbb{N}. \end{cases} \tag{3}$$

System (1) is considered as the driver system. By similar arguments, the corresponding response systems with control input is given as

$$\begin{cases} \dot{\tilde{z}}_p(t) = -d_p \tilde{z}_p(t) + \sum_{q=1}^{n} h^*_{pq}(\tilde{z}_p(t)) f_q(\tilde{z}_q(t)) + \sum_{q=1}^{n} w^*_{pq}(\tilde{z}_p(t)) g_q(\tilde{z}_q(t-\tau_q(t))) + I_p + u_p(t), \quad t \neq t_l, \\ \tilde{z}_p(t_l) = (1+\eta_l) \tilde{z}_p(t_l^-), t = t_l, l \in \mathbb{N}, \end{cases} \tag{4}$$

where $u_p(t)$ be the designed fixed-time synchronization control input to be designed later and the initial condition is denoted as $\tilde{z}_p(s) = \psi_p(s)$, $s \in [-\tau, 0]$. Assume $e_p(t) = \tilde{z}_p(t) - z_p(t)$ be the synchronization error and thus the synchronization error system can be described as

$$\begin{cases} \dot{e}_p(t) = -d_p e_p(t) + \sum_{q=1}^{n} h^*_{pq}(e_p(t)) f_q(e_q(t)) + \sum_{q=1}^{n} w^*_{pq}(e_p(t)) g_q(e_q(t-\tau_q(t))) + u_p(t), \quad t \neq t_l, \\ e_p(t_l) = (1+\eta_l) e_p(t_l^-), t = t_l, l \in \mathbb{N}, \end{cases} \tag{5}$$

and where the initial conditions of error system is assumed to be $\varphi_p(s) = \psi_p(s) - \phi_p(s)$ for $s \in [-\tau, 0]$. $h^*_{pq}(e_p(t)) f_q(e_q(t)) = h^*_{pq}(\tilde{z}_p(t)) f_q(\tilde{z}_q(t)) - h^*_{pq}(z_p(t)) f_q(z_q(t))$, $w^*_{pq}(e_p(t)) g_q(e_q(t-\tau_q(t))) = w^*_{pq}(\tilde{z}_p(t)) g_q(\tilde{z}_q(t-\tau_q(t))) - w^*_{pq}(z_p(t)) g_q(z_q(t-\tau_q(t)))$.

To obtain the main results, we make following assumptions throughout this paper.

Assumption 1. Suppose that the activation functions f_q and g_q are Lipschitz continuous. That is, for any $q = 1, 2, \dots, n$, there exist real numbers L_q and M_q such that

$$|f_q(u) - f_q(v)| \le L_q |u - v|, \quad |g_q(u) - g_q(v)| \le M_q |u - v|,$$

for all $u, v \in \mathbb{R}$, $u \neq v$.

Assumption 2. Suppose that the activation functions f_q and g_q are bounded. That is, for any $q = 1, 2, \dots, n$, there exist real numbers R_q and S_q such that

$$|f_q(u)| \le R_q, \quad |g_q(u)| \le S_q,$$

for all $u \in \mathbb{R}$.

Assumption 3. The impulse sequence $\{t_l, l \in \mathbb{N}\}$ belongs to $\Omega(\tau_{min}, \tau_{max}) \triangleq \{t_0 < t_1 < \cdots < t_l < \cdots, \lim_{t_l\to+\infty} = +\infty,\ \tau_{min} \le t_l - t_{l-1} \le \tau_{max}\}$, where τ_{min}, τ_{max} are positive constants satisfying $\tau_{min} \le \tau_{max}$.

Let $\omega(t, s)$ be the number of impulsive jumps on the time interval (s, t). Suppose that there exist a positive constant ρ such that

$$\frac{t-s}{\tau_{max}} - \rho \le \omega(t, s) \le \frac{t-s}{\tau_{min}} + \rho.$$

Lemma 1 ([31]). *Suppose $W : \mathbb{R}^n \to \mathbb{R}^+$ be a radially unbounded and continuous function which satisfies the following two conditions:*

$$1.\, W(e(t)) = 0 \Longleftrightarrow e(t) = 0,$$
$$2.\, W(e(t)) \leq -aW^{\alpha}(e(t)) - bW^{\beta}(e(t)) - cW(e(t)),$$

where $a, b, c > 0$, $\alpha > 1$ and $0 < \beta < 1$.

Then the origin of error system (5) is fixed-time stable, and the settling-time $T(e_0)$ is bounded by

$$T(e_0) \leq T_{max} = \frac{1}{c(1-\beta)} \ln(1 + \frac{c}{b}) + \frac{1}{c(\alpha - 1)} \ln(1 + \frac{c}{a}) \text{ for } \forall e_0 \in \mathbb{R}^n.$$

Lemma 2 ([45]). *If $z_1, z_2, \ldots, z_n \geq 0$, $0 < q_1 \leq 1$ and $q_2 > 1$, then*

$$\sum_{p=1}^{n} z_p^{q_1} \geq \left(\sum_{p=1}^{n} z_p\right)^{q_1}, \quad \sum_{p=1}^{n} z_p^{q_2} \geq n^{1-q_2} \left(\sum_{p=1}^{n} z_p\right)^{q_2}.$$

Lemma 3 ([19]). *Suppose that Assumptions 1 and 2 hold, then for any $u, v \in \mathbb{R}^n$, $p, q \in \mathbb{N}$, following inequalities hold*

$$|K(h_{pq}(v))f_q(v_q) - K(h_{pq}(u))f_q(u_q)| \leq \vec{h}_{pq} L_q |v_q - u_q| + R_q |\hat{h}_{pq} - \check{h}_{pq}|,$$
$$|K(w_{pq}(v))f_q(v_q) - K(w_{pq}(u))f_q(u_q)| \leq \vec{w}_{pq} M_q |v_q - u_q| + S_q |\hat{w}_{pq} - \check{w}_{pq}|,$$

where $\vec{h}_{pq} = \max\{|\hat{h}_{pq}|, |\check{h}_{pq}|\}$, $\vec{w}_{pq} = \max\{|\hat{w}_{pq}|, |\check{w}_{pq}|\}$.

Definition 1 ([46]). The derive-response systems (1) and (4) are said to be finite-time synchronized, if for any initial value e_0, there eixst a constant $T(e_0) > 0$ and $e(t) = (e_1(t), e_2(t), \ldots, e_n(t))^T$, such that $\lim_{t \to T(e_0)} \|e(t)\| = 0$, and $\|e(t)\| = 0$ for $\forall t \geq T(e_0)$, where $T(e_0)$ is called the settling-time.

Definition 2 ([46]). The derive-response systems (1) and (4) are said to be fixed-time synchronized, if the following two conditions hold:

(1) The system (4) can synchronize with the system (1) in finite-time.

(2) The settling-time function $T(e_0)$ is bounded for any initial value e_0, i.e., there exists a fixed constant $T_{max} > 0$ such that $T(e_0) \leq T_{max}$.

Definition 3 ([47]). The Dini's upper right-hand derivative of a continuous function $V(t) : \mathbb{R} \to \mathbb{R}$ is defined by

$$D^+(V(t)) = \lim_{h \to 0^+} \sup \frac{V(t+h) - V(t)}{h}.$$

3. Main results

In order to establish some sufficient conditions to achieve the fixed-time synchronization between the drive-response systems (1) and (4), we design the fixed-time controller $u_p(t)$ as follows

$$u_p(t) = -\sigma_p e_p(t) - \text{sign}(e_p(t)) \left[\varrho_p + l_1 |e_p(t)|^{\alpha_1} + l_2 |e_p(t)|^{\alpha_2} + \varsigma_p |e_p(t - \tau_p(t))|\right], \tag{6}$$

where σ_p, ϱ_p, ς_p are positive parameters to be designed later, l_1, l_2 are some positive constants, and the real number α_1, α_2 satisfy $\alpha_1 > 1$ and $0 < \alpha_2 < 1$.

Theorem 1. *Under Assumptions 1, 2 and 3, if the control gain parameters σ_p, ϱ_p and ς_p satisfy the following conditions:*

$$\begin{cases} \sigma_p \geq -d_p + \sum_{q=1}^{n} \vec{h}_{qp} L_p, \\ \varsigma_p \geq \sum_{q=1}^{n} \vec{w}_{qp} M_p, \\ \varrho_p \geq \sum_{q=1}^{n} (R_q |\hat{h}_{pq} - \check{h}_{pq}| + S_q |\hat{w}_{pq} - \check{w}_{pq}|), \end{cases} \tag{7}$$

for $p, q = 1, 2, \ldots, n$, then the drive-response systems (1) and (4) *can achieve FXTS under the controller* (6)*. Furthermore, the settling-time is estimated as* *(i)* $\widetilde{T} = T_1^* + T_2^*$, *when* $0 < \mu_l \triangleq |1+\eta_l| < 1 \ \forall\ l \in \mathbb{N}$, *where* $T_1^* = \frac{\ln\left[1+\frac{\mu_l^{\rho(1-\alpha_1)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{max}}\right)}{\bar{l}_1}\right]}{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{max}}\right)(\alpha_1-1)}$ *and* $T_2^* = \frac{\ln\left[\frac{\bar{l}_2\mu_l^{\rho(1-\alpha_2)}}{\mu_l^{-\rho(1-\alpha_2)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)+\bar{l}_2\mu_l^{\rho(1-\alpha_2)}}\right]}{\left(\frac{\ln\mu_l}{\tau_{max}}-\bar{l}_3\right)(1-\alpha_2)}$.

(ii) $\breve{T} = \frac{\ln\left[1+\frac{\bar{l}_3}{\bar{l}_1}\right]}{\bar{l}_3(\alpha_1-1)} - \frac{\ln\left[\frac{\bar{l}_2}{\bar{l}_2+\bar{l}_3}\right]}{\bar{l}_3(1-\alpha_2)}$, *when* $\mu_l = 1\ \forall\ l \in \mathbb{N}$.

Proof. Construct the Lyapunov function

$$W(t) = \sum_{p=1}^{n} |e_p(t)|, \tag{8}$$

for $t \neq t_k$, and thus the derivative of $W(t)$ along the trajectory of the error system (5) can be calculated as

$$\begin{aligned}
D^+(W(t)) &= \sum_{p=1}^{n} \mathrm{sign}(e_p(t))\dot{e}_p(t),\\
&= \sum_{p=1}^{n} \mathrm{sign}(e_p(t))\left[-d_pe_p(t) + \sum_{q=1}^{n} h_{pq}^*(e_p(t))f_q(e_q(t)) + \sum_{q=1}^{n} w_{pq}^*(e_p(t))g_q(e_q(t-\tau_q(t))) + u_p(t)\right]\\
&\le -\sum_{p=1}^{n} d_p|e_p(t)| + \sum_{p=1}^{n}\sum_{q=1}^{n} |h_{pq}^*(e_p(t))f_q(e_q(t))| + \sum_{p=1}^{n}\sum_{q=1}^{n} |w_{pq}^*(e_p(t))g_q(e_q(t-\tau_q(t)))|\\
&+ \sum_{p=1}^{n} \mathrm{sign}(e_p(t))\left[-\sigma_pe_p(t) - \mathrm{sign}(e_p(t))\left(\varrho_p + l_1|e_p(t)|^{\alpha_1} + l_2|e_p(t)|^{\alpha_2} + \varsigma_p|e_p(t-\tau_p(t))|\right)\right]\\
&\le -\sum_{p=1}^{n} d_p|e_p(t)| + \sum_{p=1}^{n}\sum_{q=1}^{n} |h_{pq}^*(e_p(t))f_q(e_q(t))| + \sum_{p=1}^{n}\sum_{q=1}^{n} |w_{pq}^*(e_p(t))g_q(e_q(t-\tau_q(t)))|\\
&- \sum_{p=1}^{n} \sigma_p|e_p(t)| - \sum_{p=1}^{n} \varrho_p - l_1\sum_{p=1}^{n} |e_p(t)|^{\alpha_1} - l_2\sum_{p=1}^{n} |e_p(t)|^{\alpha_2} - \sum_{p=1}^{n} \varsigma_p|e_p(t-\tau_p(t))|.
\end{aligned} \tag{9}$$

According to Lemma 3, one has

$$\begin{cases} |h_{pq}^*(e_p(t))f_q(e_q(t))| \le \vec{h}_{pq}L_q|e_q(t)| + R_q|\hat{h}_{pq} - \check{h}_{pq}|,\\ |w_{pq}^*(e_p(t))g_q(e_q(t-\tau_q(t)))| \le \vec{w}_{pq}M_q|e_q(t-\tau_q(t))| + S_q|\hat{w}_{pq} - \check{w}_{pq}|. \end{cases} \tag{10}$$

From (9) and (10), we have

$$\begin{aligned}
D^+(W(t)) &\le \sum_{p=1}^{n}(-d_p-\sigma_p)|e_p(t)| + \sum_{p=1}^{n}\sum_{q=1}^{n} \vec{h}_{pq}L_q|e_q(t)| + \sum_{p=1}^{n}\sum_{q=1}^{n} \vec{w}_{pq}M_q|e_q(t-\tau_q(t))| + \sum_{p=1}^{n}\sum_{q=1}^{n} R_q|\hat{h}_{pq}-\check{h}_{pq}|\\
&+ \sum_{p=1}^{n}\sum_{q=1}^{n} S_q|\hat{w}_{pq}-\check{w}_{pq}| - \sum_{p=1}^{n}\varrho_p - l_1\sum_{p=1}^{n}|e_p(t)|^{\alpha_1} - l_2\sum_{p=1}^{n}|e_p(t)|^{\alpha_2} - \sum_{p=1}^{n}\varsigma_p|e_p(t-\tau_p(t))|\\
&\le \sum_{p=1}^{n}\left(-d_p-\sigma_p+\sum_{q=1}^{n}\vec{h}_{qp}L_p\right)|e_p(t)| + \sum_{p=1}^{n}\left(\sum_{q=1}^{n}\vec{w}_{qp}M_p - \varsigma_p\right)|e_p(t-\tau_p(t))|\\
&+ \sum_{p=1}^{n}\left[\sum_{q=1}^{n}(R_q|\hat{h}_{pq}-\check{h}_{pq}| + S_q|\hat{w}_{pq}-\check{w}_{pq}|) - \varrho_p\right] - l_1\sum_{p=1}^{n}|e_p(t)|^{\alpha_1} - l_2\sum_{p=1}^{n}|e_p(t)|^{\alpha_2}.
\end{aligned} \tag{11}$$

If $\sigma_p \geq -d_p + \sum_{q=1}^{n} \vec{h}_{qp} L_p$, $\varsigma_p \geq \sum_{q=1}^{n} \vec{w}_{qp} M_p$, $\varrho_p \geq \sum_{q=1}^{n} (R_q |\hat{h}_{pq} - \check{h}_{pq}| + S_q |\hat{w}_{pq} - \check{w}_{pq}|)$, $p, q = 1, 2, \ldots, n$, one can obtain

$$D^+(W(t)) \leq -l_1 \sum_{p=1}^{n} |e_p(t)|^{\alpha_1} - l_2 \sum_{p=1}^{n} |e_p(t)|^{\alpha_2} - \sum_{p=1}^{n} c_p |e_p(t)|, \tag{12}$$

where $c_p = d_p + \sigma_p - \sum_{q=1}^{n} \vec{h}_{qp} L_p > 0$.

Let $l_3 = \min_p \{c_p\}$, then we have

$$D^+(W(t)) \leq -l_1 \sum_{p=1}^{n} |e_p(t)|^{\alpha_1} - l_2 \sum_{p=1}^{n} |e_p(t)|^{\alpha_2} - l_3 \sum_{p=1}^{n} |e_p(t)|. \tag{13}$$

In view of Lemma 2, we obtain

$$\sum_{p=1}^{n} |e_p(t)|^{\alpha_1} \geq n^{1-\alpha_1} \left(\sum_{p=1}^{n} |e_p(t)| \right)^{\alpha_1}, \quad \sum_{p=1}^{n} |e_p(t)|^{\alpha_2} \geq \left(\sum_{p=1}^{n} |e_p(t)| \right)^{\alpha_2}. \tag{14}$$

Now, (13) and (14) yield

$$\begin{aligned} D^+(W(t)) &\leq -l_1 n^{1-\alpha_1} \left(\sum_{p=1}^{n} |e_p(t)| \right)^{\alpha_1} - l_2 \left(\sum_{p=1}^{n} |e_p(t)| \right)^{\alpha_2} - l_3 \sum_{p=1}^{n} |e_p(t)| \\ &= -\bar{l}_1 W^{\alpha_1}(t) - \bar{l}_2 W^{\alpha_2}(t) - \bar{l}_3 W(t), \end{aligned} \tag{15}$$

where $\bar{l}_1 = l_1 n^{1-\alpha_1}$, $\bar{l}_2 = l_2$, $\bar{l}_3 = l_3$.

When $t = t_l$, it follows from (8) that

$$\begin{aligned} W(t_l) &= \sum_{p=1}^{n} |e_p(t_l)| = \sum_{p=1}^{n} |(1+\eta_l) e_p(t_l^-)| \\ &\leq |1+\eta_l| W(t_l^-) = \mu_l W(t_l^-), \end{aligned} \tag{16}$$

where $\mu_l = |1+\eta_l|$, and we take $0 < \mu_l \leq 1$.

Further, we construct the following comparison system to study the fixed-time synchronization of error system (5) under the controller (6), we have

$$\begin{cases} \dot{V}(t) = \begin{cases} -\bar{l}_1 V^{\alpha_1}(t) - \bar{l}_3 V(t), V(t) \geq 1, t \neq t_l, \\ -\bar{l}_2 V^{\alpha_2}(t) - \bar{l}_3 V(t), 0 < V(t) < 1, t \neq t_l, \\ 0, \ \ t \neq t_l, \end{cases} \\ V(t_l) = \mu_l V(t_l^-), t = t_l, \\ V(0) = V_0 = \sum_{p=1}^{n} |e_p(0)|. \end{cases} \tag{17}$$

From (15), (16) with (17), we get the inequality $0 \leq W(t) \leq V(t)$. That is, if there exists $\widetilde{T} > 0$, for $t \geq \widetilde{T}$ such that $V(t) \equiv 0$, then $W(t) \equiv 0$ for $t \geq \widetilde{T}$. Therefore, to prove that system (5) is fixed-time stable, we need to show that the zero solution of system (17) is fixed-time stable.

When $V(t) \geq 1$, we assume the transformation $\Psi(t) = V^{1-\alpha_1}(t)$. Since $\alpha_1 > 1$ and from (17), we get that $\Psi(t) \to 0^+$ as $V(t) \to +\infty$ and $\Psi(t) \to 1^+$ as $V(t) \to 1^+$. Therefore

$$\begin{cases} \dot{\Psi}(t) = \bar{l}_1(\alpha_1 - 1) + \bar{l}_3(\alpha_1 - 1)\Psi(t), t \neq t_l, \quad 0 < \Psi(t) \leq 1, \\ \Psi(t_l) = \bar{\mu}_l \Psi(t_l^-), t = t_l, \\ \Psi(0) = V_0^{1-\alpha_1}, \end{cases} \tag{18}$$

where $\overline{\mu}_l = \mu_l^{1-\alpha_1}$. It is not difficult to get that $V(t) \to 1^+$ is equivalent to $\Psi(t) \to 1^-$. Let us take the transformation $\Psi(t) = V^{1-\alpha_2}(t)$, where $0 < \alpha_2 < 1$ for $0 < V(t) < 1$, so that we have

$$\begin{cases} \dot{\Psi}(t) = -\bar{l}_2(1-\alpha_2) - \bar{l}_3(1-\alpha_2)\Psi(t), t \neq t_l, \quad 0 \leq \Psi(t) < 1, \\ \Psi(t_l) = \tilde{\mu}_l \Psi(t_l^-), t = t_l, \\ \Psi(0) = 1, \end{cases} \tag{19}$$

where $\tilde{\mu}_l = \mu_l^{1-\alpha_2}$. It follows that $V(t) \to 0^+$ is equivalent to $\Psi(t) \to 0^+$.

The fixed-time stability of (17) is decomposed into the following two processes: (i) the trajectories of (18) approach 1 in a fixed-time T_1^*; (ii) the trajectories of (19) approach 0 from 1 in a fixed-time T_2^*. Then we can get that $V(t) \to 0$ in the fixed-time $\widetilde{T} = T_1^* + T_2^*$ for any initial value V_0. Next we divide the proof into following two cases: $0 < \mu_l < 1$ and $\mu_l = 1$.

Case 1: $0 < \mu_l < 1$, then, $\bar{\mu}_l > 1$ and $0 < \tilde{\mu} < 1$. Thus, we have the condition $V(t_l^-) \geq 1$ and $V(t_l) \geq 1$. Now, from (18) and from the formula of variation of parameters, we have

$$\Psi(t) = e^{\bar{l}_3(\alpha_1-1)t} \bar{\mu}_l^{\omega(t,0)} \Psi(0) + \bar{l}_1(\alpha_1 - 1) \int_0^t e^{\bar{l}_3(\alpha_1-1)(t-s)} \bar{\mu}_l^{\omega(t,s)} ds. \tag{20}$$

Since $\Psi(0) = \bar{\mu}_l^{\omega(0,0)} \Psi(0) < 1$ and $\lim_{t\to\infty} \Psi(t) = \infty$, there exists T_1^* such that $\lim_{t\to T_1^*} \Psi(t) = 1$ and $0 < \Psi(t) < 1$ for $0 < t < T_1^*$. According to Eq. (20), we have

$$e^{\bar{l}_3(\alpha_1-1)t} \bar{\mu}_l^{\omega(t,0)} \Psi(0) + \bar{l}_1(\alpha_1 - 1) \int_0^t e^{\bar{l}_3(\alpha_1-1)(t-s)} \bar{\mu}_l^{\omega(t,s)} ds = 1,$$

which implies

$$\bar{l}_1(\alpha_1 - 1) \int_0^t e^{\bar{l}_3(\alpha_1-1)(t-s)} \bar{\mu}_l^{\omega(t,s)} ds \leq 1.$$

It follows from Assumption 3 that $\bar{\mu}_l^{\frac{t-s}{\tau_{max}}-\rho} \leq \bar{\mu}_l^{\omega(t,s)} \leq \bar{\mu}_l^{\frac{t-s}{\tau_{min}}+\rho}$ so that

$$e^{\bar{l}_3(\alpha_1-1)t} \int_0^t e^{-\bar{l}_3(\alpha_1-1)s} \bar{\mu}_l^{\frac{t-s}{\tau_{max}}-\rho} ds \leq \frac{1}{\bar{l}_1(\alpha_1 - 1)}.$$

After solving the above equation, we obtain

$$t \leq \frac{\ln\left[1 + \frac{\bar{\mu}_l^{\rho}\left(\bar{l}_3(\alpha_1-1) + \frac{\ln \bar{\mu}_l}{\tau_{max}}\right)}{\bar{l}_1(\alpha_1-1)}\right]}{\left(\bar{l}_3(\alpha_1 - 1) + \frac{\ln \bar{\mu}_l}{\tau_{max}}\right)} = T_1^*.$$

Substituting $\bar{\mu}_l = \mu_l^{1-\alpha_1}$ into the above inequality, we have

$$T_1^* = \frac{\ln\left[1 + \frac{\mu_l^{\rho(1-\alpha_1)}\left(\bar{l}_3 - \frac{\ln \mu_l}{\tau_{max}}\right)}{\bar{l}_1}\right]}{\left(\bar{l}_3 - \frac{\ln \mu_l}{\tau_{max}}\right)(\alpha_1 - 1)}. \tag{21}$$

Similar to above analysis, we estimate the fixed-time T_2^* such that system (19) approach 0 from 1. From (20), we find that

$$\Psi(t) = e^{-\bar{l}_3(1-\alpha_2)t} \tilde{\mu}_l^{\omega(t,0)} \Psi(0) - \bar{l}_2(1 - \alpha_2) \int_0^t e^{-\bar{l}_3(1-\alpha_2)(t-s)} \tilde{\mu}_l^{\omega(t,s)}. \tag{22}$$

Since $0 < \tilde{\mu}_l < 1$, from Assumption 3 we have $\tilde{\mu}_l^{\frac{t-s}{\tau_{min}}+\rho} \leq \tilde{\mu}_l^{\omega(t,s)} \leq \tilde{\mu}_l^{\frac{t-s}{\tau_{max}}-\rho}$. Now, considering the fixed-time T_2^* for $\Psi(t)$ changes from 1 to 0, we know that, $\Psi(0) = 1$ and $\Psi(T_2^*) = 0$. From (22), we therefore have

$$
\begin{aligned}
\Psi(t) &= e^{-\bar{l}_3(1-\alpha_2)t}\tilde{\mu}_l^{\omega(t,0)} - \bar{l}_2(1-\alpha_2)\int_0^t e^{-\bar{l}_3(1-\alpha_2)(t-s)}\tilde{\mu}_l^{\omega(t,s)}ds,\\
&\le e^{-\bar{l}_3(1-\alpha_2)t}\tilde{\mu}_l^{\frac{t}{\tau_{max}}-\rho} - \bar{l}_2(1-\alpha_2)e^{-\bar{l}_3(1-\alpha_2)t}\int_0^t e^{\bar{l}_3(1-\alpha_2)s}\tilde{\mu}_l^{\frac{t-s}{\tau_{min}}+\rho}ds\\
&= e^{-\bar{l}_3(1-\alpha_2)t}\tilde{\mu}_l^{\frac{t}{\tau_{max}}-\rho} - \bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}\left[\frac{1-e^{-\left[\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}\right]t}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}}\right]\\
&\le \left[\tilde{\mu}_l^{-\rho} + \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}}\right]e^{-\left[\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}\right]t} - \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}} \triangleq H(t).
\end{aligned}
$$

Since $H(0)=\tilde{\mu}_l^{-\rho}>0$, $H(+\infty)=-\frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}}<0$, we have $\dot{H}(t)<0$, and there exists unique T_2^* satisfying $H(T_2^*)=0$, so that we can get

$$
H(T_2^*) = \left[\tilde{\mu}_l^{-\rho} + \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}}\right]e^{-\left[\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}\right]t} - \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}} = 0,
$$

which implies that

$$
T_2^* = \frac{\ln\left[\frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}}{\tilde{\mu}_l^{-\rho}\left(\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}\right)+\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{\rho}}\right]}{\left(\frac{\ln\tilde{\mu}_l}{\tau_{max}}-\bar{l}_3\right)(1-\alpha_2)}.
$$

Substituting $\tilde{\mu}_l=\mu_l^{1-\alpha_2}$ into the above equation leads to

$$
T_2^* = \frac{\ln\left[\frac{\bar{l}_2\mu_l^{\rho(1-\alpha_2)}}{\mu_l^{-\rho(1-\alpha_2)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)+\bar{l}_2\mu_l^{\rho(1-\alpha_2)}}\right]}{\left(\frac{\ln\mu_l}{\tau_{max}}-\bar{l}_3\right)(1-\alpha_2)}. \tag{23}
$$

Thus, when it comes to the second condition $V(t_l^-)\ge 1$ and $V(t_l)<1$. This implies that the time interval of $V(t)>1$ converges to 1 is obviously smaller than T_1^* and the time interval of $0<V(t)<1$ from 1 converges to 0 is obviously smaller than T_2^*. Therefore, we may conclude that the error system (5) can achieve stability in fixed-time $T_1^*+T_2^*$. In others words, when $0<\mu_l<1$ system (4) and (1) can be synchronized within the fixed-time $\widetilde{T}=T_1^*+T_2^*$, where

$$
\widetilde{T} = \frac{\ln\left[1+\frac{\mu_l^{\rho(1-\alpha_1)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{max}}\right)}{\bar{l}_1}\right]}{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{max}}\right)(\alpha_1-1)} + \frac{\ln\left[\frac{\bar{l}_2\mu_l^{\rho(1-\alpha_2)}}{\mu_l^{-\rho(1-\alpha_2)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)+\bar{l}_2\mu_l^{\rho(1-\alpha_2)}}\right]}{\left(\frac{\ln\mu_l}{\tau_{max}}-\bar{l}_3\right)(1-\alpha_2)}. \tag{24}
$$

Case 2: $\mu_l=1$. Then it can be obtained that $\tilde{\mu}_l=\bar{\mu}_l=1$ and using the same formula as (20), one can get

$$
\Psi(t) = e^{\bar{l}_3(\alpha_1-1)t}\Psi(0)+\bar{l}_1(\alpha_1-1)e^{\bar{l}_3(\alpha_1-1)t}\int_0^t e^{-\bar{l}_3(\alpha_1-1)s}ds.
$$

Thus, we have

$$
e^{\bar{l}_3(\alpha_1-1)t}\int_0^t e^{-\bar{l}_3(\alpha_1-1)s}ds \le \frac{1}{\bar{l}_1(\alpha_1-1)},
$$

$$
t \le \frac{\ln\left[1+\frac{\bar{l}_3}{\bar{l}_1}\right]}{\bar{l}_3(\alpha_1-1)} = T_1^*.
$$

Since $\Psi(0)=1$, $\Psi(T_2^*)=0$ and $\tilde{\mu_l}=1$, one can obtain from (22) that

$$\Psi(t)=e^{-\bar{l}_3(1-\alpha_2)t}-\bar{l}_2(1-\alpha_2)e^{-\bar{l}_3(1-\alpha_2)t}\int_0^t e^{\bar{l}_3(1-\alpha_2)s}ds,$$

$$\Psi(T_2^*)=e^{-\bar{l}_3(1-\alpha_2)t}-\bar{l}_2(1-\alpha_2)e^{-\bar{l}_3(1-\alpha_2)t}\int_0^t e^{\bar{l}_3(1-\alpha_2)s}ds=0.$$

Solving the above equation gives

$$T_2^*=\frac{\ln\left[\frac{\bar{l}_2}{\bar{l}_2+\bar{l}_3}\right]}{-\bar{l}_3(1-\alpha_2)}.$$

Hence, when $\mu_l=1$, the drive-response system (1) and (4) can be synchronized within the fixed-time $\check{T}$ given by

$$\check{T}=T_1^*+T_2^*=\frac{\ln\left[1+\frac{\bar{l}_3}{\bar{l}_1}\right]}{\bar{l}_3(\alpha_1-1)}-\frac{\ln\left[\frac{\bar{l}_2}{\bar{l}_2+\bar{l}_3}\right]}{\bar{l}_3(1-\alpha_2)}. \tag{25}$$

Therefore, the proof is completed.

Remark 1. Theorem 1 establishes a novel FXTS on MNNs constituting time-varying delay and synchronizing impulsive effects. As the continuous part of Lyapunov inequality in the above mentioned theorem, defined as $\dot{W}(e(t))\le -aW^{\alpha}(e(t))-bW^{\beta}(e(t))-cW(e(t))$, constitutes one more term than in the previous papers [37], where the continuous part of Lyapunov inequality is given as $\dot{W}(e(t))\le -aW^{\alpha}(e(t))-bW^{\beta}(e(t))$, the convergence rate of the system (5) under Theorem 1 is faster than that of [37]. Also, a less conservative and high precisive estimate of settling-time is provided in Theorem 1. The above mentioned statements are numerically verified in the numerical Section 5 of the article.

Remark 2. It is well known that, impulse are of two types: synchronizing impulses and desynchronizing impulses. One can easily notice that the impulse used in Theorem 1 is of synchronizing kind, which is beneficial for synchronization of systems. Certainly, a question arises, what happens if we consider desynchronizing impulses! Are the systems (1) and (4) can be synchronized in fixed-time in that case too! The answer of this query is YES, which is verified by the next theorem, Theorem 2.

Theorem 2. *Under Assumptions 1, 2 and 3 to hold, if the parameters σ_p, ϱ_p and ς_p satisfy Eq. (7) in Theorem 1 and the following inequality holds:*

$$\bar{l}_3>\frac{\ln\mu_l}{\tau_{min}}, \tag{26}$$

then the drive-response system (1) and *(4) can be achieved FXTS under the controllers (6) for desynchronizing impulsive effects. Furthermore, the settling-time is estimated as $\widehat{T}=T_1^*+T_2^*$, when $0<\mu_l\triangleq|1+\eta_l|>1\ \forall\ l\in\mathbb{N}$, where $T_1^*=\frac{\ln\left[1+\frac{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)}{\bar{l}_1\mu_l^{\rho(1-\alpha_1)}}\right]}{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)(\alpha_1-1)}$ and $T_2^*=\frac{\ln\left[\frac{\bar{l}_2\mu_l^{-\rho(1-\alpha_2)}}{\mu_l^{\rho(1-\alpha_2)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{max}}\right)+\bar{l}_2\mu^{-\rho(1-\alpha_2)}}\right]}{\left(\frac{\ln\mu_l}{\tau_{min}}-\bar{l}_3\right)(1-\alpha_2)}$.*

Proof. The proof of this theorem for $t\neq t_l$ is similar as the proof of Theorem 1, so we omitted this part here.

Now, when $t=t_l$, it follows from (8) that

$$\begin{aligned}W(t_l)&=\sum_{p=1}^{n}|e_p(t_l)|=\sum_{p=1}^{n}|(1+\eta_l)e_p(t_l^-)|\\&\le|1+\eta_l|W(t_l^-)=\mu_lW(t_l^-),\end{aligned} \tag{27}$$

where $\mu_l = |1+\eta_l|$, and we take $\mu_l > 1$. We constructed the following comparison system to study the fixed-time synchronization of error system (5) under the controller (6) for desynchronizing impulsive effects

$$\begin{cases} \dot{V}(t) = \begin{cases} -\bar{l}_1 V^{\alpha_1}(t) - \bar{l}_3 V(t), V(t) \geq 1, t \neq t_l, \\ -\bar{l}_2 V^{\alpha_2}(t) - \bar{l}_3 V(t), 0 < V(t) < 1, t \neq t_l, \\ 0, \ \ t \neq t_l, \end{cases} \\ V(t_l) = \mu_l V(t_l^-), t = t_l, \\ V(0) = \sum_{p=1}^{n} |e_p(0)|. \end{cases} \tag{28}$$

Combining (15), (27) with (28), we get the inequality $0 \leq W(t) \leq V(t)$. That is, there exist $\widehat{T} > 0$, for $t \geq \widehat{T}$ such that $V(t) \equiv 0$, then $W(t) \equiv 0$ for $t \geq \widehat{T}$. Therefore, to prove that system (5) is fixed-time stable, we need to show that the zero solution of system (28) is fixed-time stable.

When $V(t) \geq 1$, let us take the transformation $\Psi(t) = V^{1-\alpha_1}(t)$. Since $\alpha_1 > 1$ and from (28), we have $\Psi(t) \to 1^+$ as $V(t) \to 1^+$ and $\Psi(t) \to 0^+$ as $V(t) \to +\infty$. Therefore

$$\begin{cases} \dot{\Psi}(t) = \bar{l}_1(\alpha_1 - 1) + \bar{l}_3(\alpha_1 - 1)\Psi(t), t \neq t_l, \quad 0 < \Psi(t) \leq 1, \\ \Psi(t_l) = \bar{\mu}_l \Psi(t_l^-), t = t_l, \\ \Psi(0) = V_0^{1-\alpha_1}, \end{cases} \tag{29}$$

where $\bar{\mu}_l = \mu_l^{1-\alpha_1}$. It follows that $V(t) \to 1^+$ is equivalent to $\Psi(t) \to 1^+$. Now, consider the transformation $\Psi(t) = V^{1-\alpha_2}(t)$, where $0 < \alpha_2 < 1$ for $0 < V(t) < 1$. Therefore, we have

$$\begin{cases} \dot{\Psi}(t) = -\bar{l}_2(1 - \alpha_2) - \bar{l}_3(1 - \alpha_2)\Psi(t), t \neq t_l, \quad 0 \leq \Psi(t) < 1, \\ \Psi(t_l) = \tilde{\mu}_l \Psi(t_l^-), t = t_l, \\ \Psi(0) = 1, \end{cases} \tag{30}$$

where $\tilde{\mu}_l = \mu_l^{1-\alpha_2}$. It is not difficult to see that $V(t) \to 1^+$ is equivalent to $\Psi(t) \to 1^+$.

Now from the above discussion, the fixed-time stability of (28) has been transformed into the following two processes: (i) the trajectories of (29) approaches 1 from 0 in a fixed-time T_1^*; (ii) the trajectories of (30) approaches 0 from 1 in fixed-time T_2^*. Then we can find that, $V(t) \to 0$ in the fixed-time $\widehat{T} = T_1^* + T_2^*$ for any initial value V_0. We shall consider the above two cases of impulsive strength $\eta_l > 1$.

It follows from (29) and from the formula of variation of parameters, we obtain

$$\Psi(t) = e^{\bar{l}_3(\alpha_1-1)t} \bar{\mu}_l^{\omega(t,0)} \Psi(0) + \bar{l}_1(\alpha_1 - 1) \int_0^t e^{\bar{l}_3(\alpha_1-1)(t-s)} \bar{\mu}_l^{\omega(t,s)} ds.$$

Since $\Psi(0) = \bar{\mu}_l^{\omega(0,0)} \Psi(0) < 1$ and $\lim_{t\to\infty} \Psi(t) = \infty$, when $0 < \bar{\mu}_l < 1$, there exist T_1^* such that $\lim_{t \to T_1^*} \Psi(t) = 1$ and $0 < \Psi(t) < 1$ for $0 < t < T_1^*$ that is

$$e^{\bar{l}_3(\alpha_1-1)t} \bar{\mu}_l^{\omega(t,0)} \Psi(0) + \bar{l}_1(\alpha_1 - 1) \int_0^t e^{\bar{l}_3(\alpha_1-1)(t-s)} \bar{\mu}_l^{\omega(t,s)} ds = 1.$$

Furthermore

$$\bar{l}_1(\alpha_1 - 1) \int_0^t e^{\bar{l}_3(\alpha_1-1)(t-s)} \bar{\mu}_l^{\omega(t,s)} ds \leq 1,$$

According to Assumption 3, $\bar{\mu}_l^{\frac{t-s}{\tau_{min}}+\rho} \leq \bar{\mu}_l^{\omega(t,s)} \leq \bar{\mu}_l^{\frac{t-s}{\tau_{max}}-\rho}$ so that

$$e^{\bar{l}_3(\alpha_1-1)t} \int_0^t e^{-\bar{l}_3(\alpha_1-1)s} \bar{\mu}_l^{\frac{t-s}{\tau_{min}}+\rho} ds \leq \frac{1}{\bar{l}_1(\alpha_1 - 1)}.$$

After solving above equation, we obtains that

$$t \leq \frac{\ln\left[1 + \frac{\left(\bar{l}_3(\alpha_1-1) + \frac{\ln \bar{\mu}_l}{\tau_{min}}\right)}{\bar{l}_1(\alpha_1-1)\bar{\mu}_l^{\rho}}\right]}{\left(\bar{l}_3(\alpha_1 - 1) + \frac{\ln \bar{\mu}_l}{\tau_{min}}\right)} = T_1^*.$$

Substituting $\bar{\mu}_l = \mu_l^{1-\alpha_1}$, we get

$$T_1^* = \frac{\ln\left[1+\frac{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)}{\bar{l}_1\mu_l^{\rho(1-\alpha_1)}}\right]}{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)(\alpha_1-1)}. \tag{31}$$

In similar way, one has from (30) that

$$\Psi(t) = e^{-\bar{l}_3(1-\alpha_2)t}\tilde{\mu}_l^{\omega(t,0)}\Psi(0) - \bar{l}_2(1-\alpha_2)\int_0^t e^{-\bar{l}_3(1-\alpha_2)(t-s)}\tilde{\mu}_l^{\omega(t,s)}. \tag{32}$$

Since $1<\tilde{\mu}_l<\infty$ and from Assumption 3 we have $\tilde{\mu}_l^{\frac{t-s}{\tau_{max}}-\rho} \le \tilde{\mu}_l^{\omega(t,s)} \le \tilde{\mu}_l^{\frac{t-s}{\tau_{min}}+\rho}$. Now, considering the fixed-time T_2^* for $\Psi(t)$ changes from 1 to 0, we know that, $\Psi(0)=1$ and $\Psi(T_2^*)=0$. From (32) that

$$\begin{aligned}
\Psi(t) &= e^{-\bar{l}_3(1-\alpha_2)t}\tilde{\mu}_l^{\omega(t,0)} - \bar{l}_2(1-\alpha_2)\int_0^t e^{-\bar{l}_3(1-\alpha_2)(t-s)}\tilde{\mu}_l^{\omega(t,s)}ds,\\
&\le e^{-\bar{l}_3(1-\alpha_2)t}\tilde{\mu}_l^{\frac{t}{\tau_{min}}+\rho} - \bar{l}_2(1-\alpha_2)e^{-\bar{l}_3(1-\alpha_2)t}\int_0^t e^{\bar{l}_3(1-\alpha_2)s}\tilde{\mu}_l^{\frac{t-s}{\tau_{max}}-\rho}ds\\
&= e^{-\bar{l}_3(1-\alpha_2)t}\tilde{\mu}_l^{\frac{t}{\tau_{min}}+\rho} - \bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}\left[\frac{1-e^{-\left[\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}\right]t}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}}\right]\\
&\le \left[\tilde{\mu}_l^{\rho} + \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}}\right]e^{-\left[\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}\right]t} - \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}} \triangleq H(t).
\end{aligned}$$

Since $H(0)=\tilde{\mu}_l^{\rho}>0$, $H(+\infty) = -\frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}}<0$, we have $\dot{H}(t)<0$, there exists unique T_2^* satisfies $H(T_2^*)=0$, so that we can get

$$H(T_2^*) = \left[\tilde{\mu}_l^{\rho} + \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}}\right]e^{-\left[\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{min}}\right]t} - \frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}}{\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}} = 0,$$

which implies that

$$T_2^* = \frac{\ln\left[\frac{\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}}{\tilde{\mu}_l^{\rho}\left(\bar{l}_3(1-\alpha_2)-\frac{\ln\tilde{\mu}_l}{\tau_{max}}\right)+\bar{l}_2(1-\alpha_2)\tilde{\mu}_l^{-\rho}}\right]}{\left(\frac{\ln\tilde{\mu}_l}{\tau_{min}}-\bar{l}_3(1-\alpha_2)\right)}.$$

Substituting $\tilde{\mu}_l = \mu_l^{1-\alpha_2}$ into the above equation, we obtain

$$T_2^* = \frac{\ln\left[\frac{\bar{l}_2\mu_l^{-\rho(1-\alpha_2)}}{\mu_l^{\rho(1-\alpha_2)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{max}}\right)+\bar{l}_2\mu^{-\rho(1-\alpha_2)}}\right]}{\left(\frac{\ln\mu_l}{\tau_{min}}-\bar{l}_3\right)(1-\alpha_2)}. \tag{33}$$

Thus, this implies that the time interval of $V(t)>1$ converges to 1 is obviously smaller than T_1^* and the time interval of $0<V(t)<1$ from 1 converges to 0 is obviously smaller than T_2^*. Therefore, we may conclude that the error system (5) can achieve stability in fixed-time $T_1^*+T_2^*$. In other words, when $\mu_l>1$ system (1) and (4) can be synchronized within the fixed-time $\widehat{T}=T_1^*+T_2^*$ given by

$$\widehat{T} = T_1^*+T_2^* = \frac{\ln\left[1+\frac{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)}{\bar{l}_1\mu_l^{\rho(1-\alpha_1)}}\right]}{\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{min}}\right)(\alpha_1-1)} + \frac{\ln\left[\frac{\bar{l}_2\mu_l^{-\rho(1-\alpha_2)}}{\mu_l^{\rho(1-\alpha_2)}\left(\bar{l}_3-\frac{\ln\mu_l}{\tau_{max}}\right)+\bar{l}_2\mu^{-\rho(1-\alpha_2)}}\right]}{\left(\frac{\ln\mu_l}{\tau_{min}}-\bar{l}_3\right)(1-\alpha_2)}. \tag{34}$$

Therefore, the proof is completed.

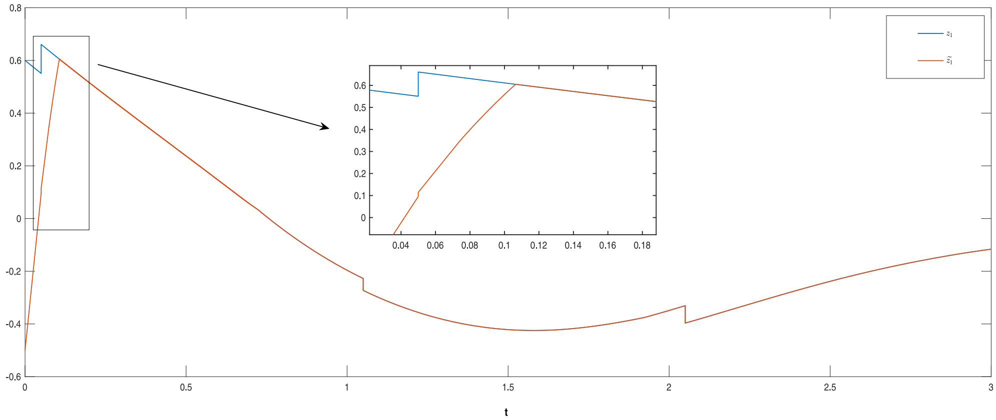

Fig. 1. Synchronization curves of z_1 and $\widetilde{z}_1$.

Remark 3. Contrary to the claim mentioned in the previous article reported in [42] that the FXTS of system (5) cannot be guaranteed when the impulsive strength $\mu_l > 1$, we have proved in Theorem 2 that if the condition $\bar{l}_3 > \frac{\ln \mu_l}{\tau_{min}}$ holds then the FXTS of MNNs can be achieved under the influence of desynchronizing impulsive effects and we can estimate the settling time $\widehat{T}$.

4. Numerical simulations and discussions

Example 1. Consider the following MNN with impulsive effects for two nodes

$$\begin{cases} \dot{z}_p(t) = -d_p z_p(t) + \sum_{q=1}^{2} h_{pq}(z_p(t)) f_q(z_q(t)) + \sum_{q=1}^{2} w_{pq}(z_p(t)) g_q(z_q(t-\tau_q(t))), \quad t \neq t_l, \\ z_p(t_l) = \mu_l z_p(t_l^-), \, t = t_l, \, l \in \mathbb{N}, \end{cases} \tag{35}$$

$p, q = 1, 2$, where $d_1 = d_2 = 1$ and the memristive connection weights are

$$h_{11}(z_1) = \begin{cases} 1.2, & |z_1| \leq 1, \\ 0.9, & |z_1| > 1, \end{cases} \quad h_{12}(z_1) = \begin{cases} -0.2, & |z_1| \leq 1, \\ -0.5, & |z_1| > 1, \end{cases}$$

$$h_{21}(z_2) = \begin{cases} -1.5, & |z_2| \leq 1, \\ -2.0, & |z_2| > 1, \end{cases} \quad h_{22}(z_2) = \begin{cases} 0.4, & |z_1| \leq 1, \\ 1.2, & |z_2| > 1, \end{cases}$$

$$w_{11}(z_1) = \begin{cases} -1.5, & |z_1| \leq 1, \\ -0.7, & |z_1| > 1, \end{cases} \quad w_{12}(z_1) = \begin{cases} -0.5, & |z_1| \leq 1, \\ 1, & |z_1| > 1, \end{cases}$$

$$w_{21}(z_2) = \begin{cases} 0.3, & |z_1| \leq 1, \\ 1.5, & |z_1| > 1, \end{cases} \quad w_{22}(z_2) = \begin{cases} -2.1, & |z_1| \leq 1, \\ -1.1, & |z_1| > 1, \end{cases}$$

The neuron activation functions and time-varying delays are defined as $f_p(z) = \tanh(z)$, $g_p(z) = \frac{1}{2}(|z+1| - |z-1|)$ and $\tau_p(t) = \frac{e^t}{1+e^t}$, $p = 1, 2$, then we have $L_p = M_p = R_p = S_p = 1$. The initial values of MNN (35) are $\phi_1(s) = 0.6$, $\phi_2(s) = 0.2$, $s \in [-1, 0]$.

The corresponding response system with impulsive effects is given by

$$\begin{cases} \dot{\widetilde{z}}_p(t) = -d_p \widetilde{z}_p(t) + \sum_{q=1}^{2} h^*_{pq}(\widetilde{z}_p(t)) f_q(\widetilde{z}_q(t)) + \sum_{q=1}^{2} w^*_{pq}(\widetilde{z}_p(t)) g_q(\widetilde{z}_q(t-\tau_q(t))) + u_p(t), \quad t \neq t_l, \\ \widetilde{z}_p(t_l) = \mu_l \widetilde{z}_p(t_l^-), \, t = t_l, \, l \in \mathbb{N}, \end{cases} \tag{36}$$

$p, q = 1, 2$, where $u_p(t)$ are the controllers. The initial values of MNN (35) are taken as $\psi_1(s) = -0.5, \psi_2(s) = 0.8$, $s \in [-1, 0]$. The evolutions of drive-response systems (1) and (4) under the controller are shown in Figs. 1 and 2

According to Theorem 1, the control gains σ_p, ϱ_p and ς_p should satisfy

$$\sigma_1 \geq -d_1 + \sum_{q=1}^{2} \vec{h}_{q1} L_1 = 2.2, \;\; \sigma_2 \geq -d_2 + \sum_{q=1}^{2} \vec{h}_{q2} L_2 = 0.7,$$

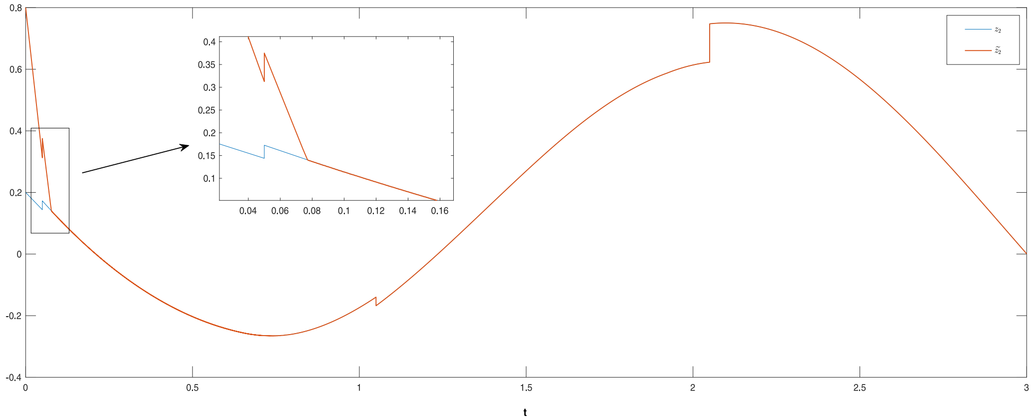

Fig. 2. Synchronization curves of z_2 and $\widetilde{z}_2$.

$$\varsigma_1 \geq \sum_{q=1}^{2} \vec{w}_{q1} M_1 = 3, \;\; \varsigma_2 \geq \sum_{q=1}^{2} \vec{w}_{q2} M_2 = 3.1,$$

$$\varrho_1 \geq \sum_{q=1}^{2} (R_q |\hat{h}_{1q} - \check{h}_{1q}| + S_q |\hat{w}_{1q} - \check{w}_{1q}|) = 2.9,$$

$$\varrho_2 \geq \sum_{q=1}^{2} (R_q |\hat{h}_{2q} - \check{h}_{2q}| + S_q |\hat{w}_{2q} - \check{w}_{2q}|) = 3.5,$$

Then we choose $\sigma_1 = 2.5$ $\sigma_2 = 1$, $\varsigma_1 = 3.1$, $\varsigma_2 = 3.2$, $\varrho_1 = 3$, $\varrho_2 = 3.6$. Furthermore, we choose $l_1 = 3$, $l_2 = 2$, $l_3 = 1$ $\alpha_1 = 1.5$, $\alpha_2 = 0.5$, then we have $\bar{l}_1 = 2.1$, $\bar{l}_2 = 2$, $\bar{l}_3 = 1$.

The impulsive strength is taken as $\mu_l = 0.8$, indicating that the impulses are synchronizing and we choose the impulsive sequences $\{t_l, l \in \mathbb{N}\}$ that satisfy $\rho = 2$, $\tau = 1$. According to Assumption 3 and [36], we can obtain that $\tau_{min} = 1.2$, $\tau_{max} = 0.8$. It can be shows that all the condition (7) of Theorem 1 are satisfied. Thus, the drive-response systems (1) and (4) can be achieved FXTS under the controllers (6) for synchronizing impulses and the settling-time can be estimated as $\widetilde{T} = 1.874$. The evolutions of the synchronization error system (5) for synchronizing impulses under the controllers (6) are shown in Fig. 3. Also it follows from Theorem 1, if the impulsive strength is taken as $\mu_l = 1$ that means the impulses are inactive and we choose the same impulsive sequences and parameters as defined above for synchronizing impulsive case, then the drive-response systems (1) and (4) can be achieved FXTS under the controllers (6) and the settling-time can be estimated as $\check{T} = 1.589$. The evolution of the synchronization error system (5) under the controller (6) for inactive impulses are shown in Fig. 4. This verifies through the above numerical simulations that the estimated settling-time in Theorem 1 is much lesser then the existing results reported in [37–39].

Example 2. Consider the same systems in Example 1, we will verify the results obtained in Theorem 2. Let $\mu_l = 1.2$ is the impulsive strength of the impulsive sequence $\{t_l, l \in \mathbb{N}\}$ with average impulsive interval $\tau_{min} = 1.2$, $\tau_{max} = 0.8$, and positive constant $\rho = 2$ such that the sufficient condition $\bar{l}_3 > \frac{\ln \mu_l}{\tau_{min}}$ is satisfied. Therefore, the drive-response systems (1) and (4) can be achieved FXTS under the controller (6) for desynchronizing impulses and the settling-time can be estimated as $\widehat{T} = 1.973$. The evolutions of the synchronization error system (5) for desynchronizing impulses under the controller (6) are shown in Fig. 5.

5. Conclusion

In this article, the problem of FXTS is studied for MNNs with time-varying delay and impulsive effects. Synchronizing and desynchronizing impulsive effects have been considered separately to investigate the MNNs to achieve the FXTS within the settling time. The results for desynchronizing impulses have not been established in any previous article reported in [37–39,42]. The main highlight of the present work is that it provides the less conservative results through numerical simulation and theoretical derivation for the case of synchronizing impulses and desynchronizing impulses. Finally, numerical examples are provided to validate the effectiveness of our proposed theoretical results. In this present article, impulsive strength is considered as a constant. If the impulsive strength can be considered as a function of time,

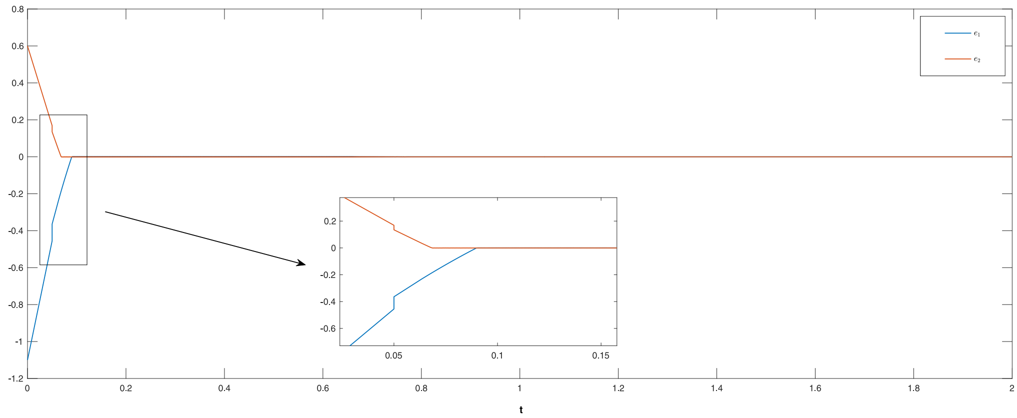

Fig. 3. The trajectories of error system (5) for synchronizing impulses ($\mu_l = 0.8$).

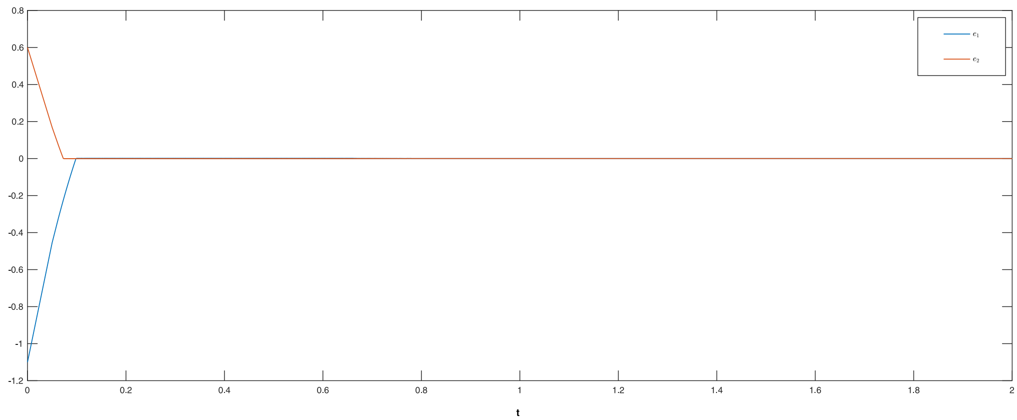

Fig. 4. The trajectories of error system (5) for inactive impulses ($\mu_l = 1$).

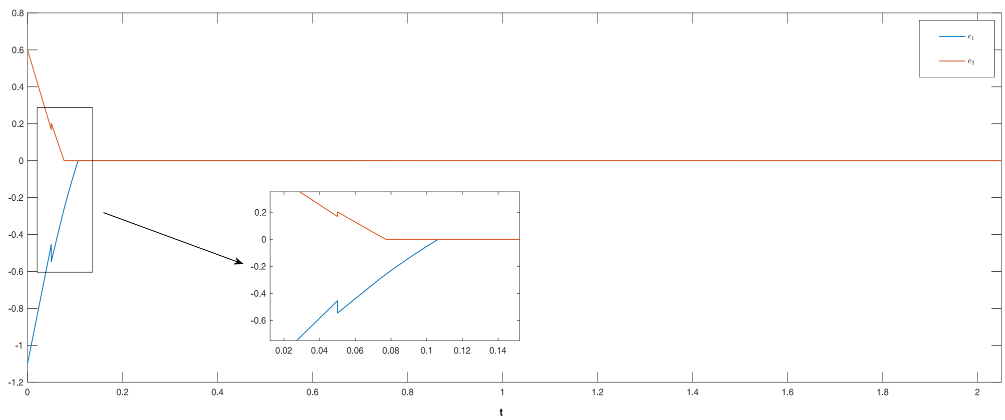

Fig. 5. The trajectories of error system (5) for desynchronizing impulses ($\mu_l = 1.2$).

which is more general, then investigating the dynamical behavior of MNNs is a more challenging task which would be thoroughly investigated in our future research work.

CRediT authorship contribution statement

Md Arzoo Jamal: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Arnab Mapui:** Methodology, Software, Writing – original draft. **Subir Das:** Methodology, Writing – original draft. **Santwana Mukhopadhyay:** Methodology, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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